

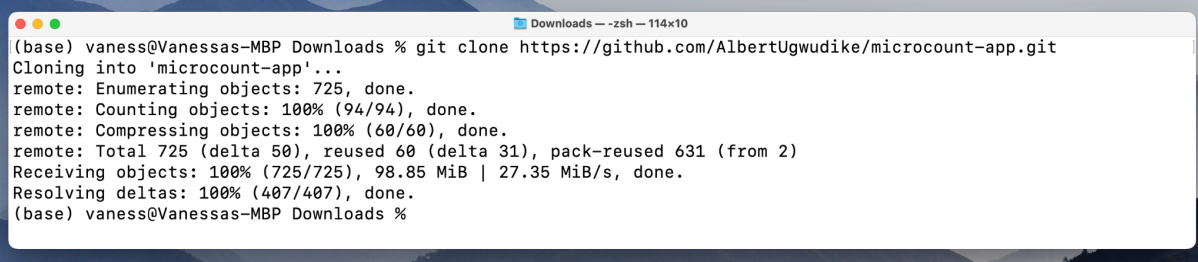
Microcount Protocol

Requirements

- MATLAB >= R2022b
- Bioformats Package (MATLAB Add-On manager)
- Parallel Computing Toolbox (MATLAB Add-On manager)
- Image Processing Toolbox (MATLAB Add-On manager)
- Java >= 21.0.2 (<https://www.oracle.com/uk/java/technologies/downloads/>)

Installation

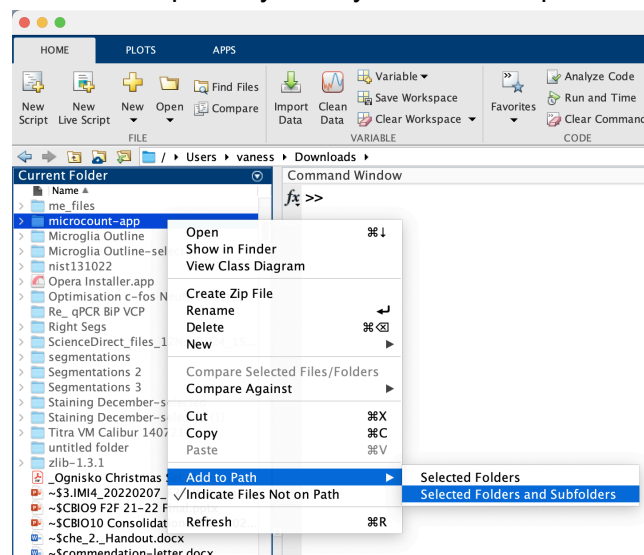
- Clone the repository
- Use command: 'git clone <https://github.com/AlbertUgwudike/microcount-app.git>'



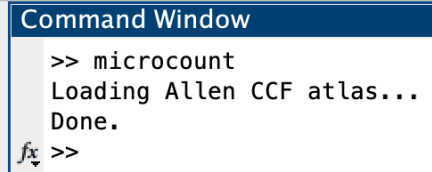
```
(base) vaness@Vanessas-MBP Downloads % git clone https://github.com/AlbertUgwudike/microcount-app.git
Cloning into 'microcount-app'...
remote: Enumerating objects: 725, done.
remote: Counting objects: 100% (94/94), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 725 (delta 50), reused 60 (delta 31), pack-reused 631 (from 2)
Receiving objects: 100% (725/725), 98.85 MiB | 27.35 MiB/s, done.
Resolving deltas: 100% (407/407), done.
(base) vaness@Vanessas-MBP Downloads %
```

Usage

- Open MATLAB
- Navigate to the downloaded repository
- Ensure the downloaded repository is on your MATLAB path

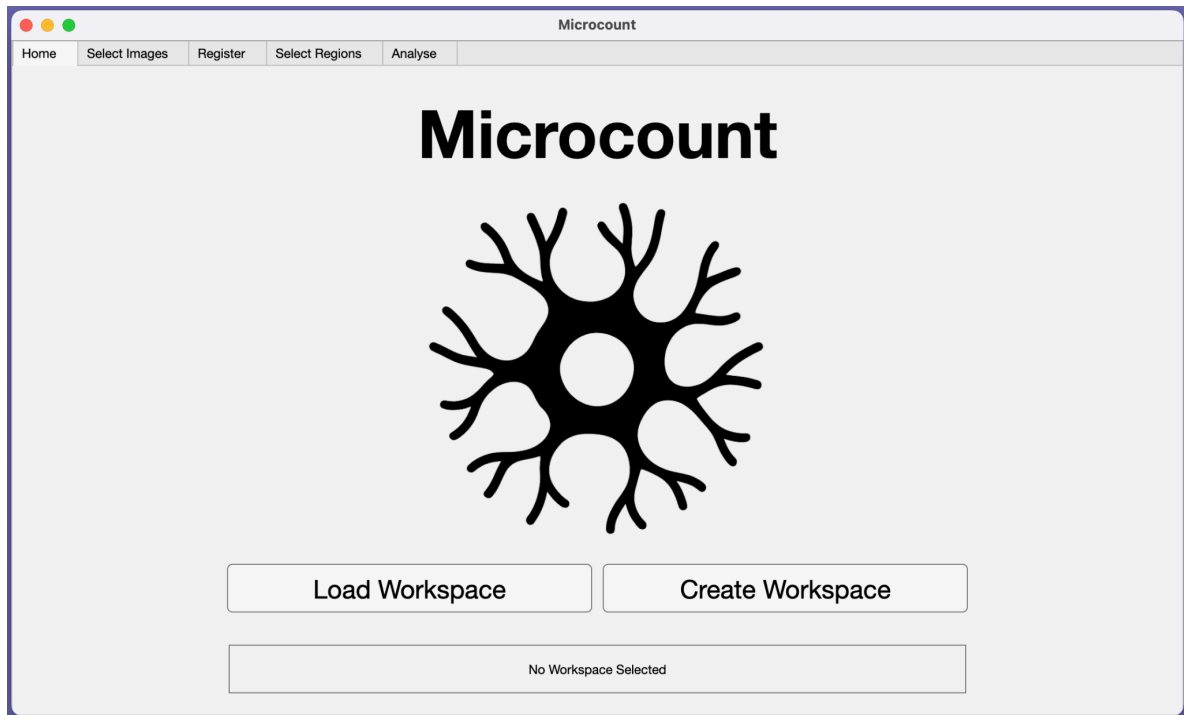


- Run the command 'microcount'



```
Command Window
>> microcount
Loading Allen CCF atlas...
Done.
fx >>
```

- The following window should appear:



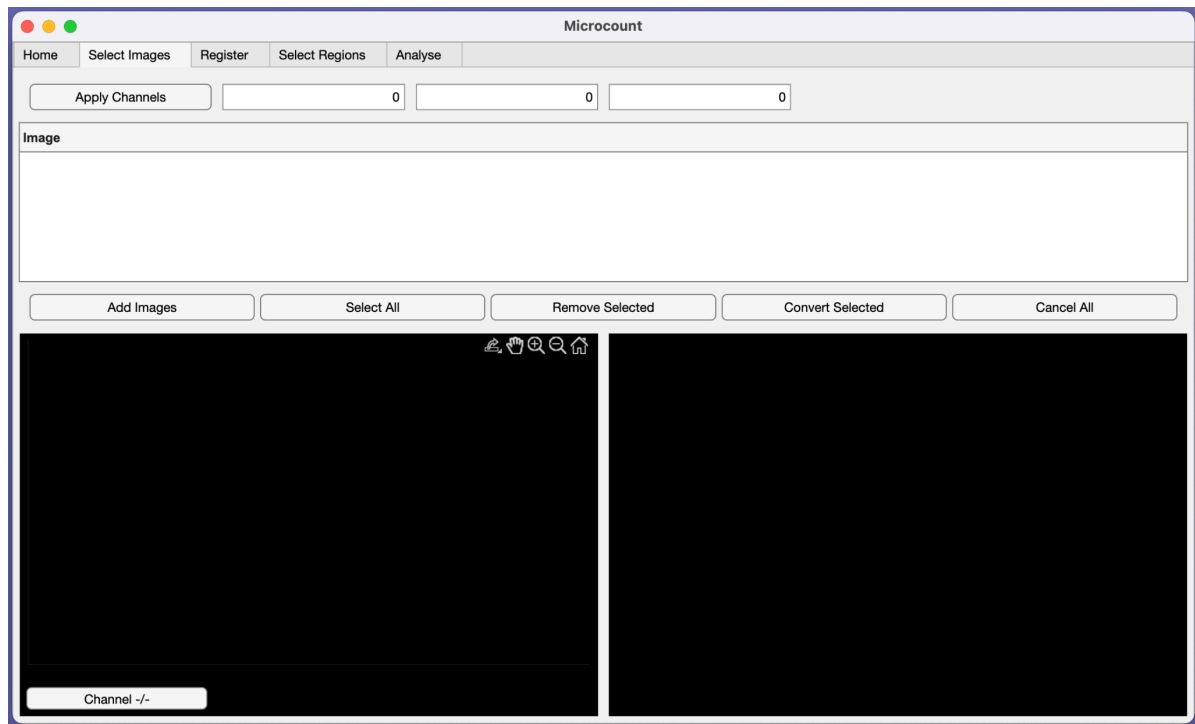
- On this tab you are able to create a new workspace or load an existing one.

You can think of a workspace as a particular set of related images, settings and results. Users may use different workspaces for different projects.

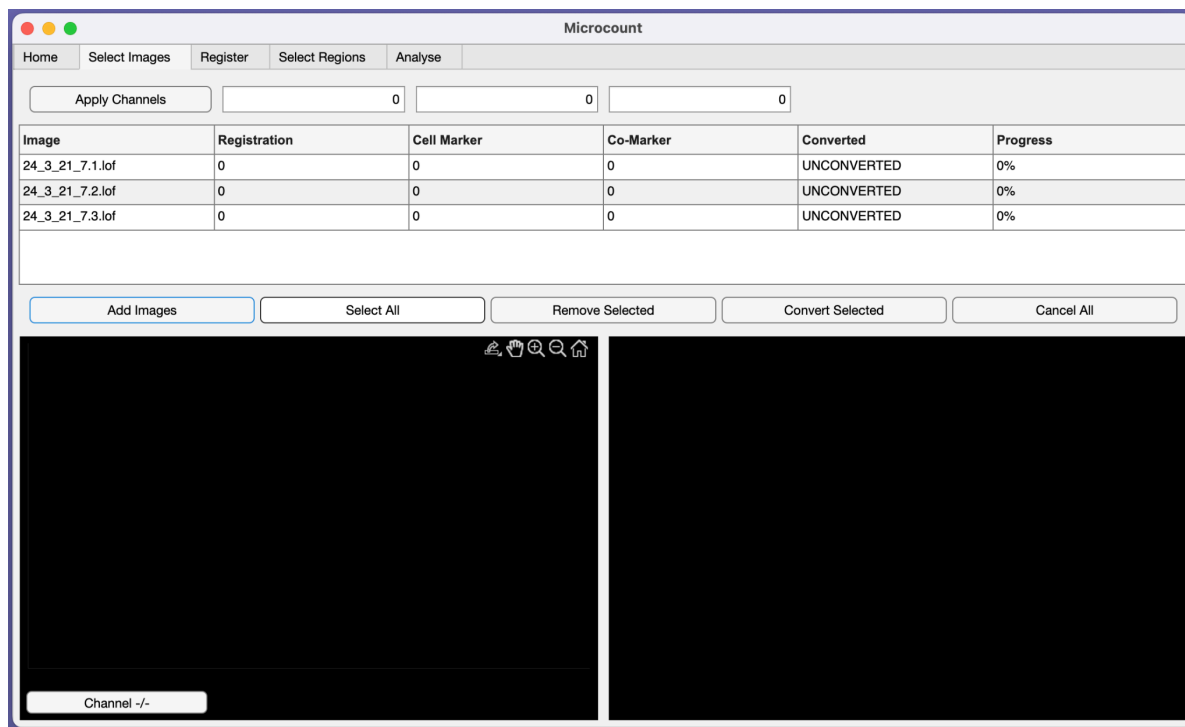
- Click 'Create Workspace' to open a window enabling you to name and create a new workspace. **Skip this step if you already have one created.**
- Then, click 'Load Workspace' to select the created workspace folder. The full path to this folder should appear beneath the buttons
- Once a workspace is loaded you may now navigate to the other sections of the pipeline

Note: Tabs are arranged in a sequential manner. If you have not selected/converted images, there will be no images to register. If there are no images registered, there will be no images for region selection etc..

- Click the 'Select Images' Tab. You will be presented with the following:



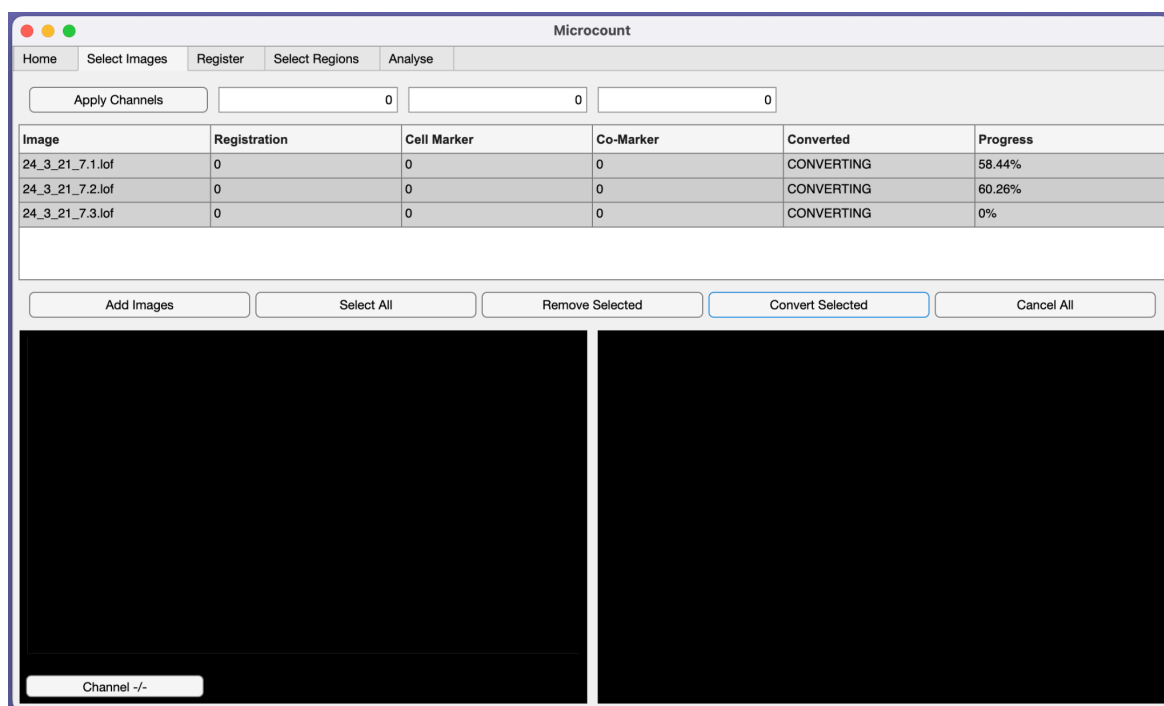
On the 'Select Images' Tab you are able to select images for further processing, additionally the channel order and specific channel names can be specified on this page as well.

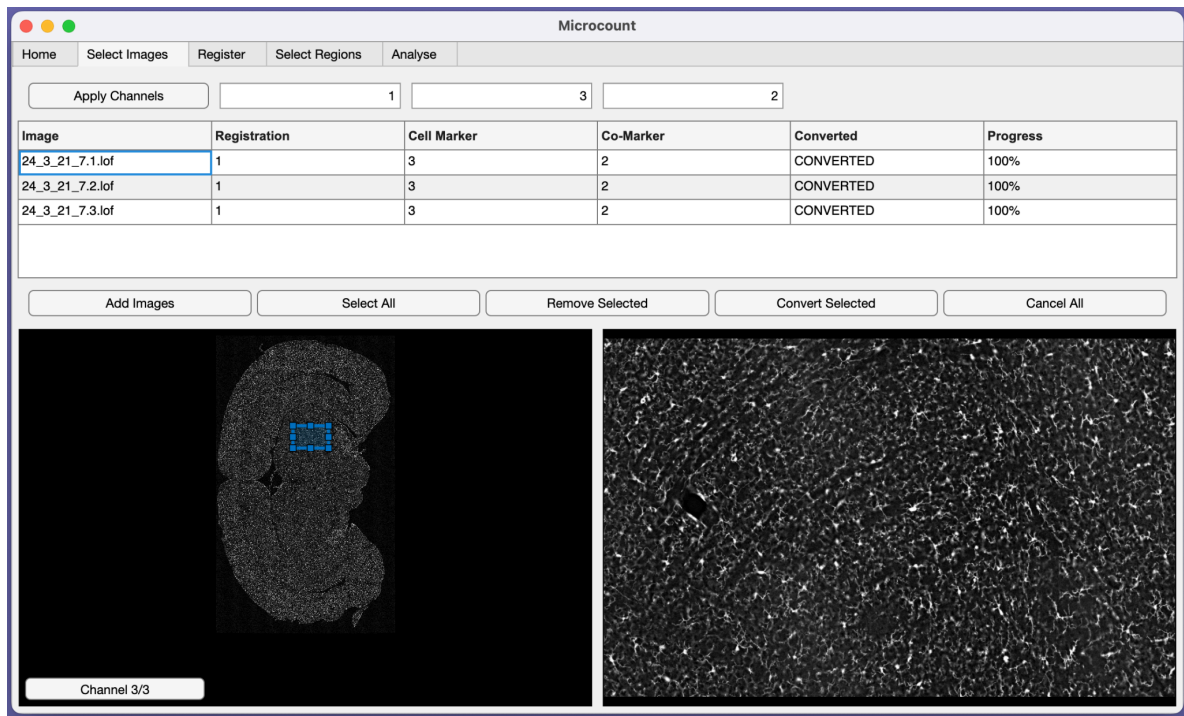


When images are first added you'll notice that 'Registration', 'Cell Marker' and 'Co-Marker' columns are all zero. These will be populated once images are converted into a more general format.

Conversion can be achieved by highlighting the desired images and clicking the 'Convert Selected' button. For large images, this can take a couple minutes. Progress of the conversion is displayed in the 'Progress Column'

*** I'm currently working on improving the memory efficiency of conversion, I recommend converting a maximum of five images at a time to avoid memory issues ***

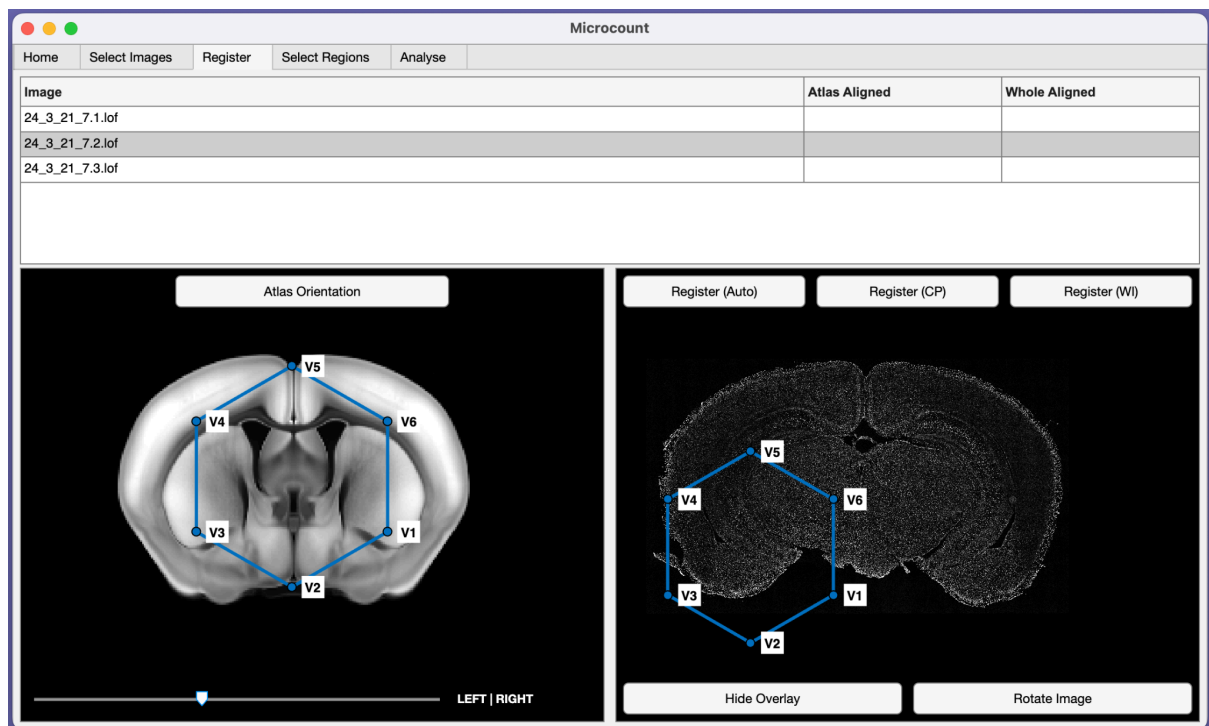




Microcount will utilise Channel 1 for registration, Channel 2 for cell segmentation (IBA1) and Channel 3 as the co-marker channel (CD68). Users are able to change the channel order for converted images. Select the images you want to configure, enter the new channel order in the corresponding fields above the table and click 'Apply Channels'.

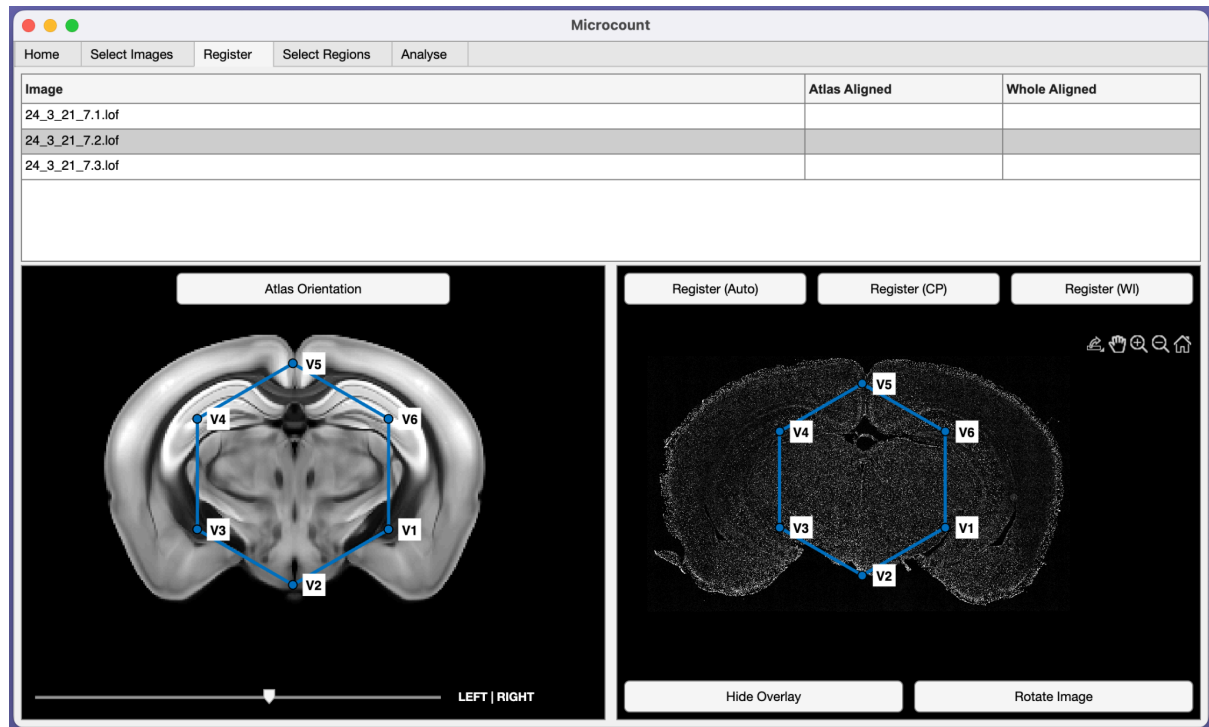
The table can also be modified directly on a per-image basis.

- Once one or more images have been converted, navigate to the 'Register Tab'
- On the register tab, images available to register will be shown in the table. Click on one of them to display its thumbnail:



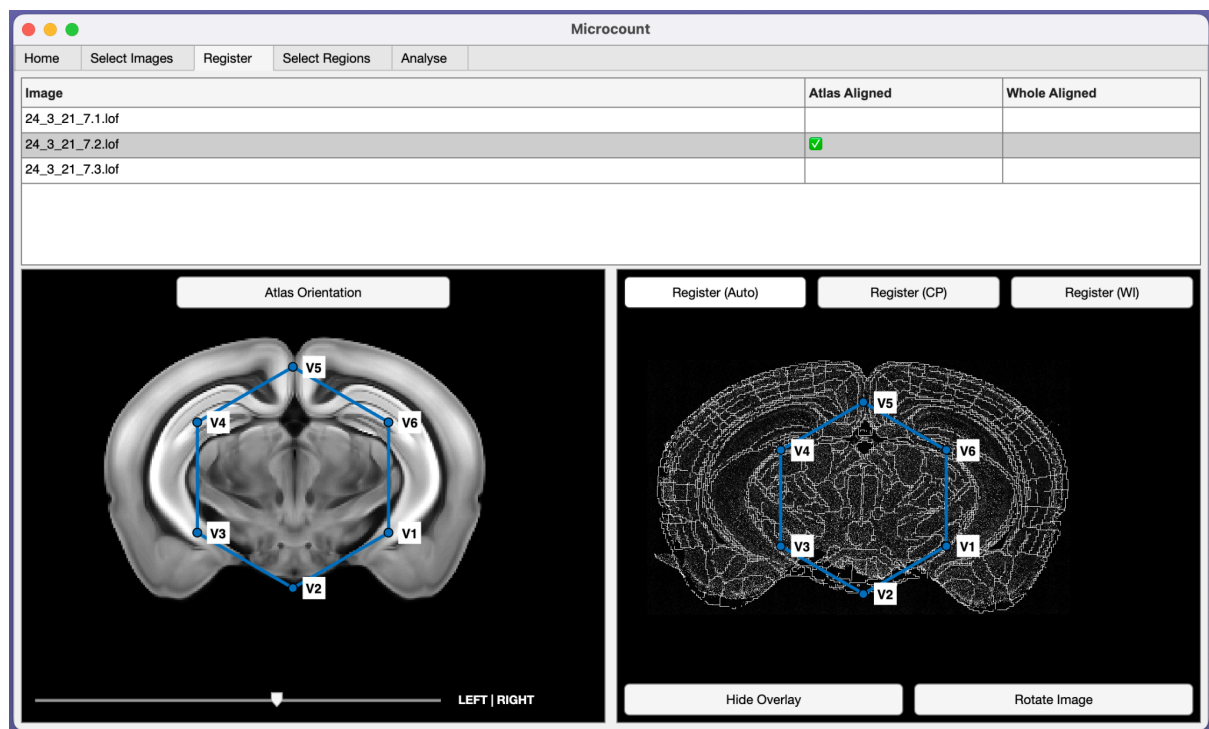
On the register tab there are various tools to assist with alignment, you are able to rotate the input image and also toggle the orientation of the atlas between axial, sagittal and coronal.

- Use the 'Rotate Image' button to rotate the input image to the correct position.
- Use the Atlas slider to choose the correct atlas depth

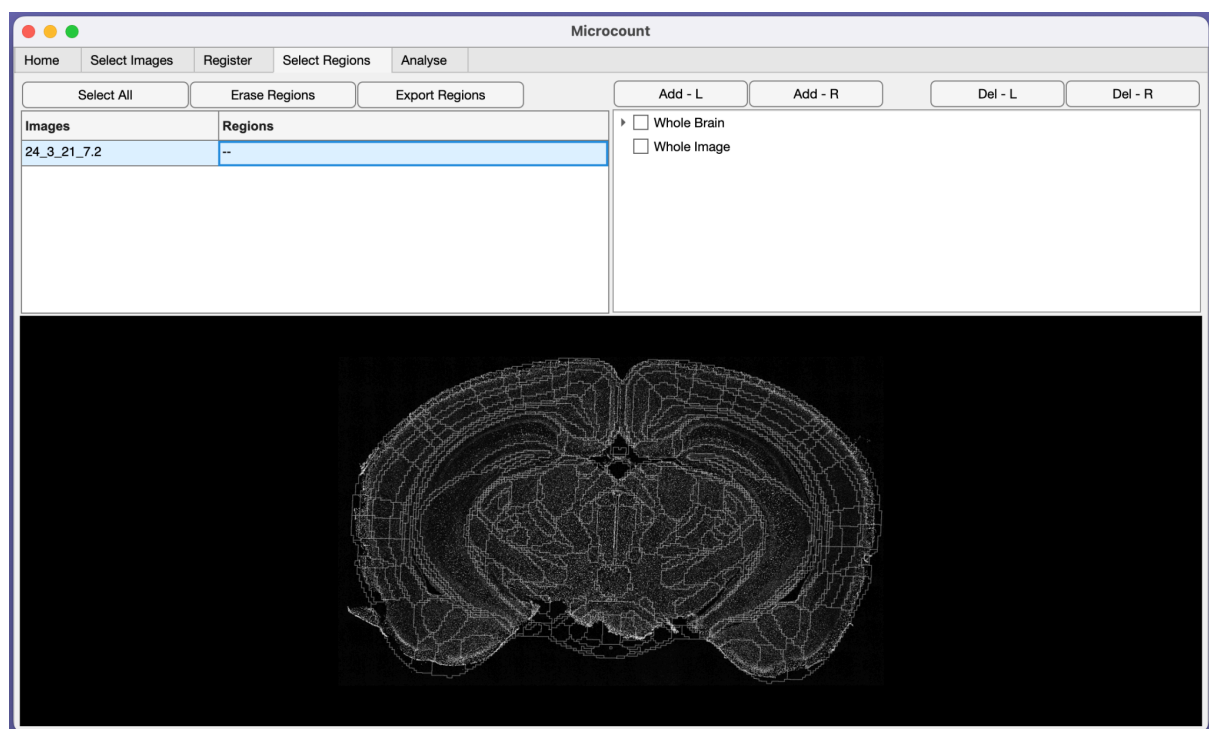


Now that both our brain section and atlas section are in similar positions we can now attempt to register. There are three options for registration that can be used 'Auto' tries to align the images based on intensity information. 'CP' attempts to align images using corresponding control points (depicted as the vertices of a blue hexagon here). 'WI' simply registers the whole image (i.e a rectangular selection covering the entire image).

- Use 'Automatic Registration' to register the image and atlas slice

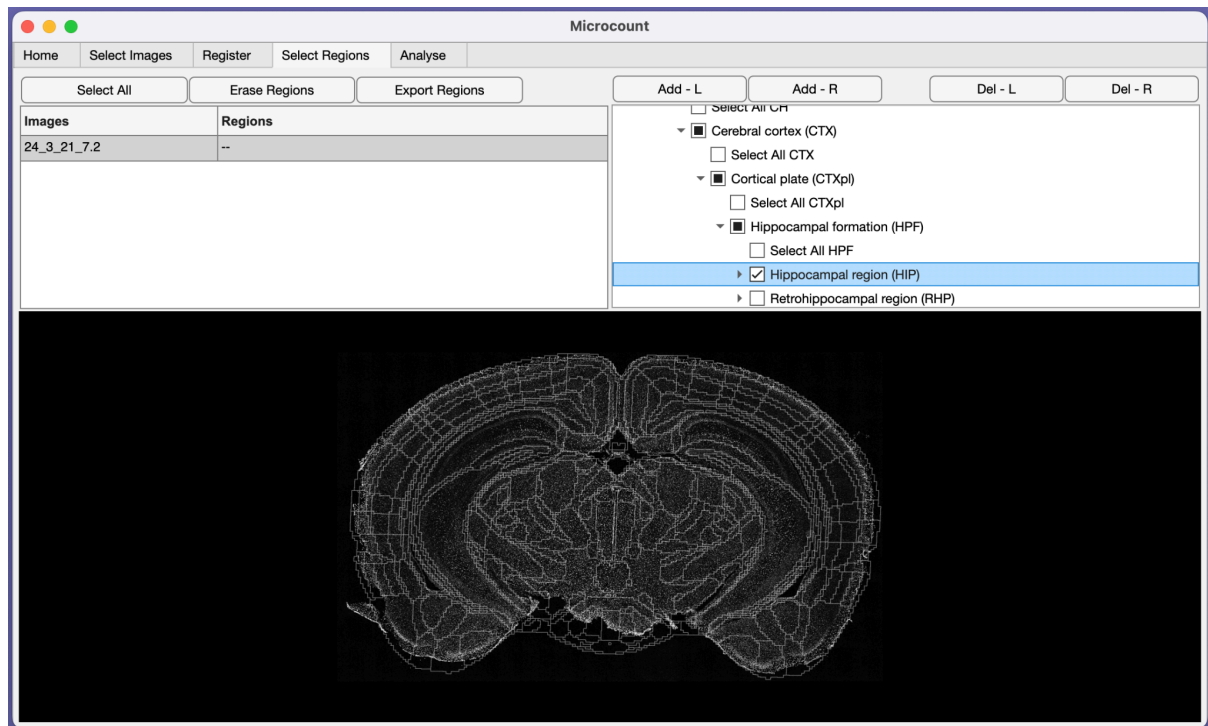


- Now that images have been registered we you may go to the Region Selection Tab
- On this tab select an image and you will be presented with the following view



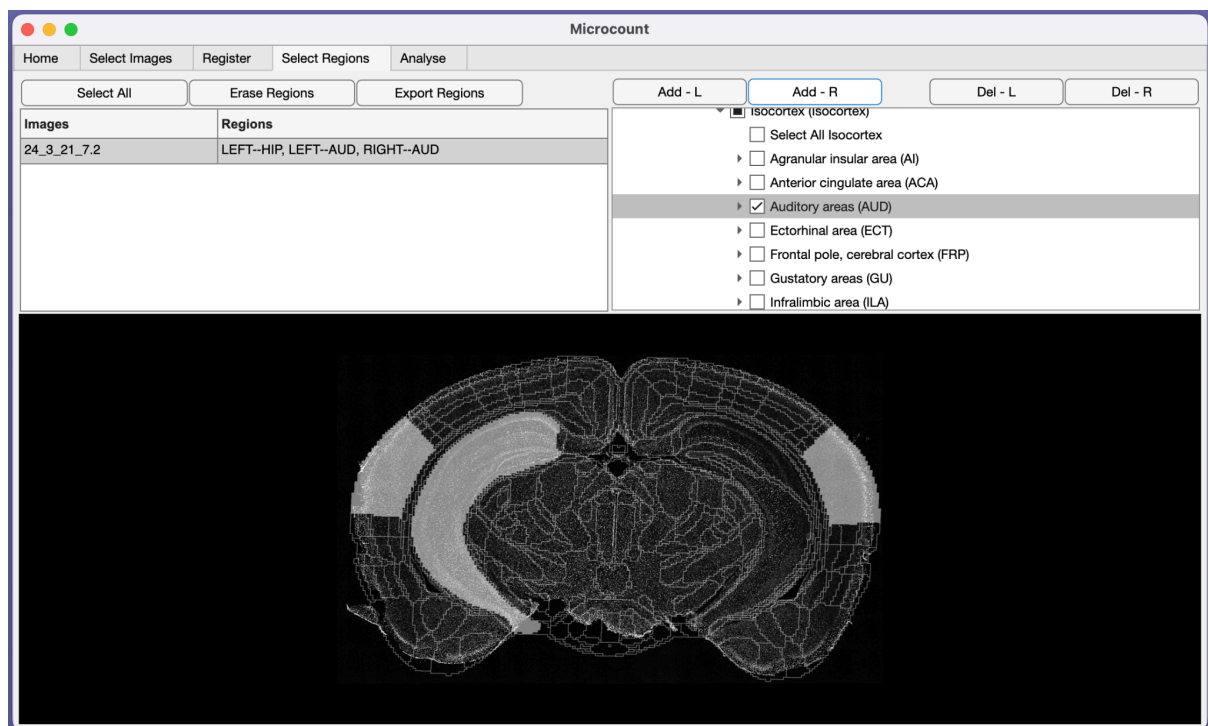
On the select regions tab, users are able to select anatomical regions of interest within each registered image for further processing. The selection of regions is achieved by navigating and selecting components of the Region Structure Tree.

- Select your anatomical regions of interest in the region selector tree



- Once selected you can add these regions to either the left or the right side of the registered mask by clicking the respective button. Regions may be removed from the left or right of selected images in a similar fashion!

NOTE: Removing regions that have already been analysed (later on) in this step effectively deletes the analysis results for that region.



Note: If a region is not present in the registered mask, it will not be added for analyses (clicking either of the 'add' buttons will do nothing in this case). Furthermore, if the registered slice is in the sagittal plane then only the left OR right buttons will appear to do anything (depending on the slice depth)

Note: On this page users are also able to export cropped full-size image regions from the original image for use elsewhere. When the 'Export Regions' button is pressed, a dialog will appear allowing you to save the selected regions to a location of your choice.

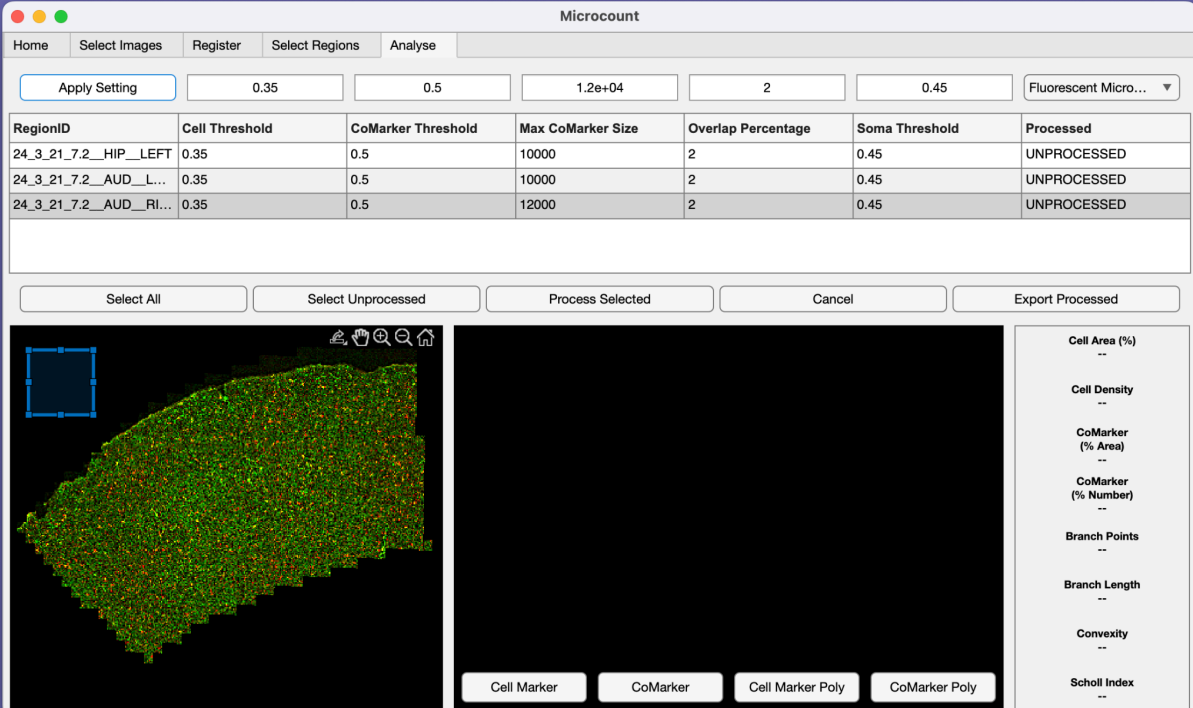
- Now that regions have been selected, you may navigate to the 'Analyse Tab'
- Click on one of the regions and you will observe the following:

RegionID	Cell Threshold	CoMarker Threshold	Max CoMarker Size	Overlap Percentage	Soma Threshold	Processed
24_3_21_7.2_HIP__LEFT	0.35	0.5	10000	2	0.45	UNPROCESSED
24_3_21_7.2_AUD__L...	0.35	0.5	10000	2	0.45	UNPROCESSED
24_3_21_7.2_AUD__RI...	0.35	0.5	10000	2	0.45	UNPROCESSED

In the 'Analyse Tab' you are able to set algorithmic parameters (to individual images by entering into the table, or to multiple images at once using the top fields and 'apply setting' button).

- Adjust the parameters passed to microcount on a per-region or whole-dataset basis as required
- **Cell Threshold:** normalised pixel values in the 'cell' channel above this threshold will be included in the mask. Microcount utilises standard normalisation so the majority of pixel values will reside between -2.5 and +2.5. **A higher cell threshold will result in a more restricted cell mask.**
- **CoMarker Threshold:** Similar to the cell threshold but for the 'co-marker' channel. **A higher cell threshold will result in a more restricted co-marker mask.**
- **Max CoMarker Size:** Often, unusually large aggregations of comarker signal are more in keeping with background signal. To mitigate this, microcount thresholds the size of co-marker 'lumps'. **A higher comarker size threshold will result in larger aggregations being included in the co-marker mask.**

- **Overlap percentage:** Often, every detected cell will colocalise with at least one co-marker pixel. For this reason microcount allows the user to set the percentage colocalisation above which a cell will be deemed activated. **A higher overlap percentage will result in fewer cells being deemed activated.**
- **Soma Threshold:** Much akin to the cell and comarker thresholds but for soma detection. **A higher soma threshold will result in a more restricted soma mask.**



Microcount

Home Select Images Register Select Regions **Analyse**

Apply Setting 0.35 0.5 1.2e+04 2 0.45 Fluorescent Micro...

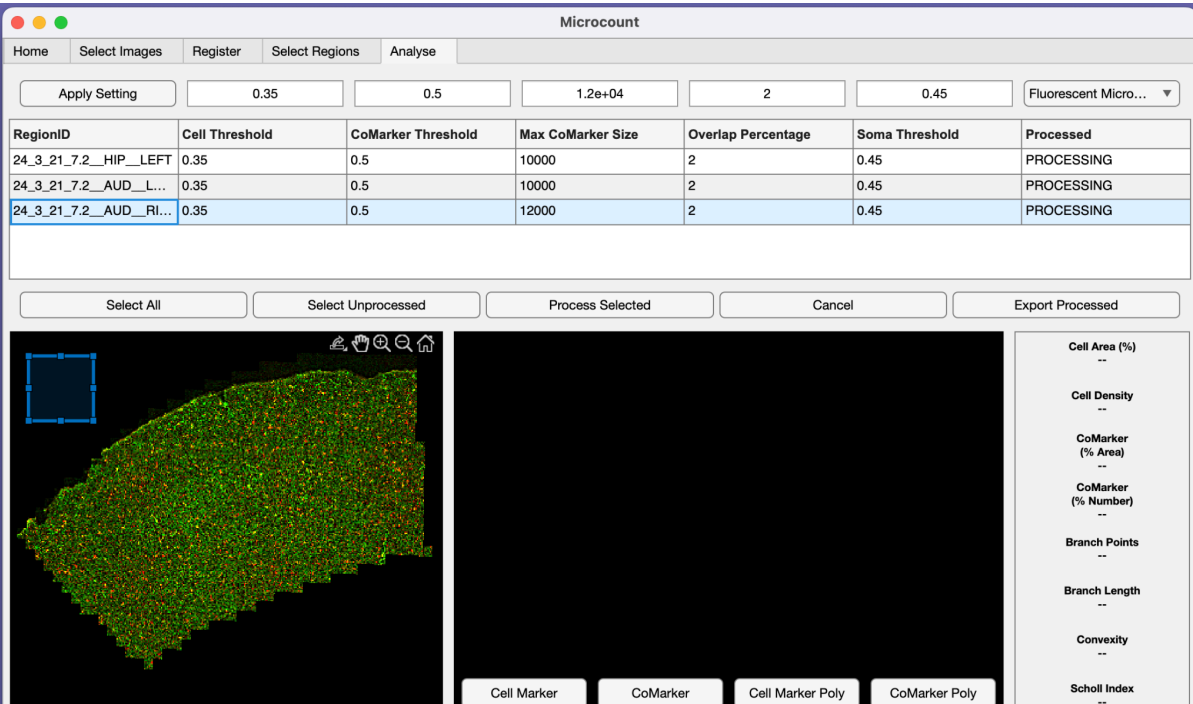
RegionID	Cell Threshold	CoMarker Threshold	Max CoMarker Size	Overlap Percentage	Soma Threshold	Processed
24_3_21_7.2_HIP_LEFT	0.35	0.5	10000	2	0.45	UNPROCESSED
24_3_21_7.2_AUD_L...	0.35	0.5	10000	2	0.45	UNPROCESSED
24_3_21_7.2_AUD_RI...	0.35	0.5	12000	2	0.45	UNPROCESSED

Select All Select Unprocessed Process Selected Cancel Export Processed

Cell Area (%) --
Cell Density --
CoMarker (% Area) --
CoMarker (% Number) --
Branch Points --
Branch Length --
Convexity --
Scholl Index --

Cell Marker CoMarker Cell Marker Poly CoMarker Poly

- Highlight the images you would like to process and click 'Process Selected' to begin analysis.



Microcount

Home Select Images Register Select Regions **Analyse**

Apply Setting 0.35 0.5 1.2e+04 2 0.45 Fluorescent Micro...

RegionID	Cell Threshold	CoMarker Threshold	Max CoMarker Size	Overlap Percentage	Soma Threshold	Processed
24_3_21_7.2_HIP_LEFT	0.35	0.5	10000	2	0.45	PROCESSING
24_3_21_7.2_AUD_L...	0.35	0.5	10000	2	0.45	PROCESSING
24_3_21_7.2_AUD_RI...	0.35	0.5	12000	2	0.45	PROCESSING

Select All Select Unprocessed Process Selected Cancel Export Processed

Cell Area (%) --
Cell Density --
CoMarker (% Area) --
CoMarker (% Number) --
Branch Points --
Branch Length --
Convexity --
Scholl Index --

Cell Marker CoMarker Cell Marker Poly CoMarker Poly

Note: For larger images this step can take quite some time, we found that for ~5Gb images, analyses would complete in a few minutes.

Note: The two main computationally intensive tasks (image conversion/downsampling and image processing) have been implemented to run in parallel in the background. Once a region is marked as complete you may proceed whilst other images are processing.

Note: Sometimes, if an image seems to be taking an unusually long amount of time to process it may be due to over-estimation of the number of cells (i.e. the soma threshold is too low). It is advised that in this circumstance, the user should cancel and re-process with a higher soma threshold (one can always fine tune this afterwards).

- Once a region has been processed, select it for viewing.
- Move the region-viewer window (blue-rectangle) to the portion of the downsampled image you would like to view.

The screenshot shows the Microcount application interface. At the top, there are tabs: Home, Select Images, Register, Select Regions, and Analyse. Below the tabs, there are input fields for various parameters: Apply Setting, 0.35, 0.5, 1.2e+04, 2, 0.45, and a dropdown menu for Fluorescent Micro... . Below these fields is a table with the following data:

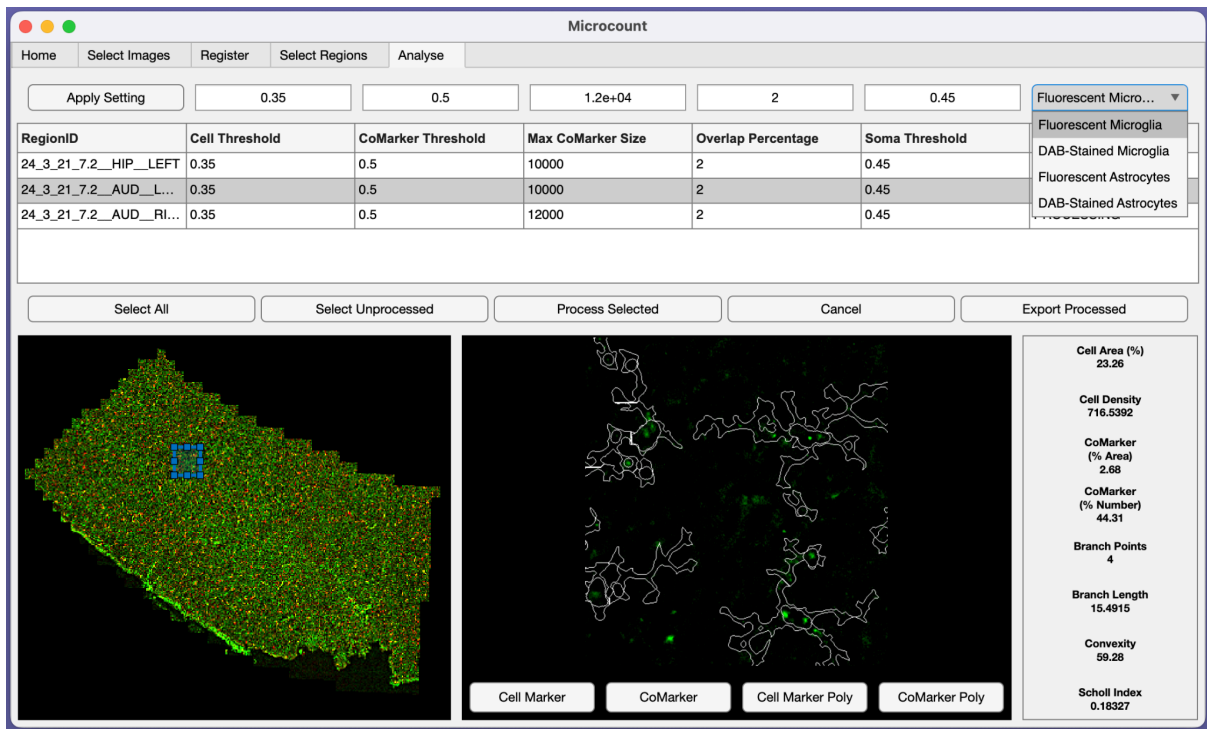
RegionID	Cell Threshold	CoMarker Threshold	Max CoMarker Size	Overlap Percentage	Soma Threshold	Processed
24_3_21_7.2_HIP__LEFT	0.35	0.5	10000	2	0.45	PROCESSING
24_3_21_7.2_AUD__L...	0.35	0.5	10000	2	0.45	PROCESSED
24_3_21_7.2_AUD__RI...	0.35	0.5	12000	2	0.45	PROCESSING

Below the table are buttons: Select All, Select Unprocessed, Process Selected, Cancel, and Export Processed. The main area of the application is divided into two panels. The left panel shows a large image with a blue rectangle indicating the region of interest. The right panel shows a zoomed-in view of the region of interest, with a blue rectangle indicating the region of interest. Below the zoomed-in view are buttons: Cell Marker, CoMarker, Cell Marker Poly, and CoMarker Poly. On the far right, there is a sidebar with various statistics:

- Cell Area (%) 23.26
- Cell Density 716.5392
- CoMarker (% Area) 2.68
- CoMarker (% Number) 44.31
- Branch Points 4
- Branch Length 15.4915
- Convexity 59.28
- Scholl Index 0.18327

Note: The region-viewer was implemented in this way due to the performance-cost of rendering full-size images within the app. If you would like to see the full image in your FOV, you can reshape the region-view window to occupy the full extent of the downsampled image.

- Adjust parameters and re-process as you deem necessary.
- Use the toggle buttons in the image viewer to hide/reveal channels or their masks



- Choose from four different algorithms dependent upon your cell type and staining. You can select the algorithm using the dropdown before processing an image.
- Once satisfied with the analysis output, you may click the 'Export Processed' button to export the data from all processed regions
- The exported data should look similar to this:

example

file_name	mask_file_name	region_ID	cell_threshold	max_comarker_size	comarker_threshold	cell_density	percentage_cell_area	percentage_comarker_area	percentage_comarker_num	average_convexity	average_soma_size	average_branch_count
/Volumes/Mor	/Volumes/MorseDrive	24_07_25-3.2	0.35	10000	0.5	285.5279	19.73	8.79	90.22	49.89	55.0967	8
/Volumes/Mor	/Volumes/MorseDrive	24_07_25-3.2	0.35	12000	0.5	282.9183	17.51	1.43	36.59	49.18	60.8645	6
/Volumes/Mor	/Volumes/MorseDrive	24_07_25-3.2	0.34	5000	0.55	220.4111	24.24	0.5	5.05	42.72	47.3869	13